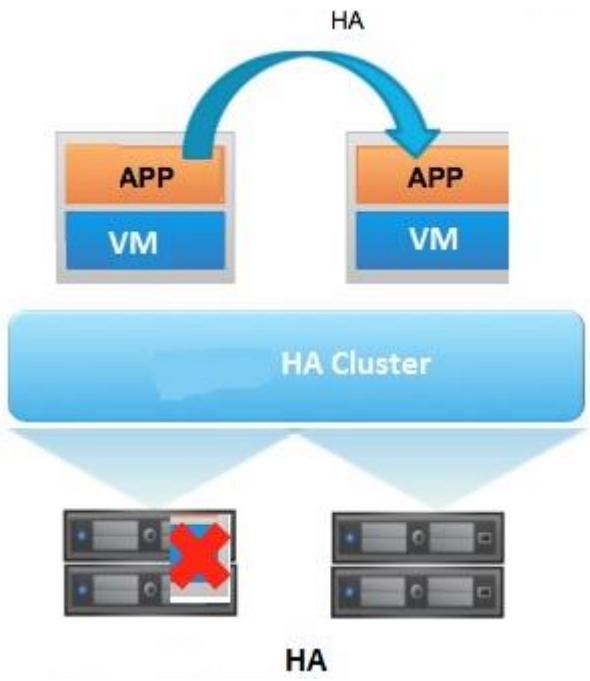


CLOUD COMPUTING COURSE

Lecture26- Virtualization Features(1)

ABSTRACT

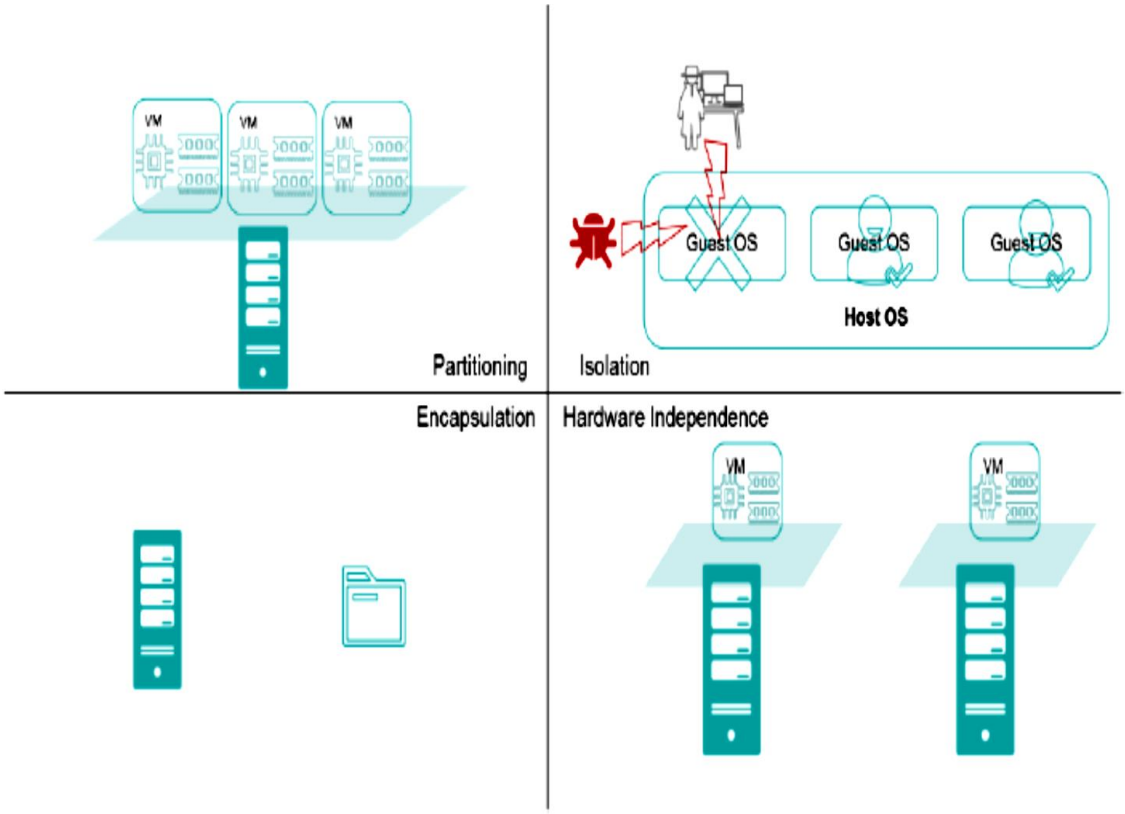
Lecture26 Introduces virtualization cluster features.

Instructor: Asma'a Khtoom

Virtualization Features

As we have discussed in previous sections, cloud computing is a service delivery model, and virtualization is a technology. Virtualization is the main technology in the Cloud Computing 1.0 era. So, what features does virtualization provide?

Before introducing virtualization features, let us go over the characteristics of virtualization again.



Based on these characteristics, we introduce virtualization features from the aspects of **clusters** and **VMs.**

Virtualization Cluster Features

A cluster is a group of computers that collectively provide resources for users, including compute, storage, and network. A complete cluster must contain all these types of resources.

1. HA (High Availability)

The cluster technology is used to overcome the limitations of a single physical host, thereby preventing service interruption or reducing system downtime.

HA in virtualization is actually the compute-layer HA, that is, **VM HA**. If a compute node becomes faulty, the other compute node in the cluster can be automatically started.

How HA is accomplished?

A virtualization cluster usually uses shared storage. A VM consists of configuration files and data disks. The data disks are stored on shared storage, and the configuration files are stored on the compute nodes. If a compute node becomes faulty, the virtualization management system (such as vCenter and VRM) rebuilds the fault VM on other nodes based on the recorded VM configuration information.

Addressed Issues

During HA, the following issues need to be addressed:

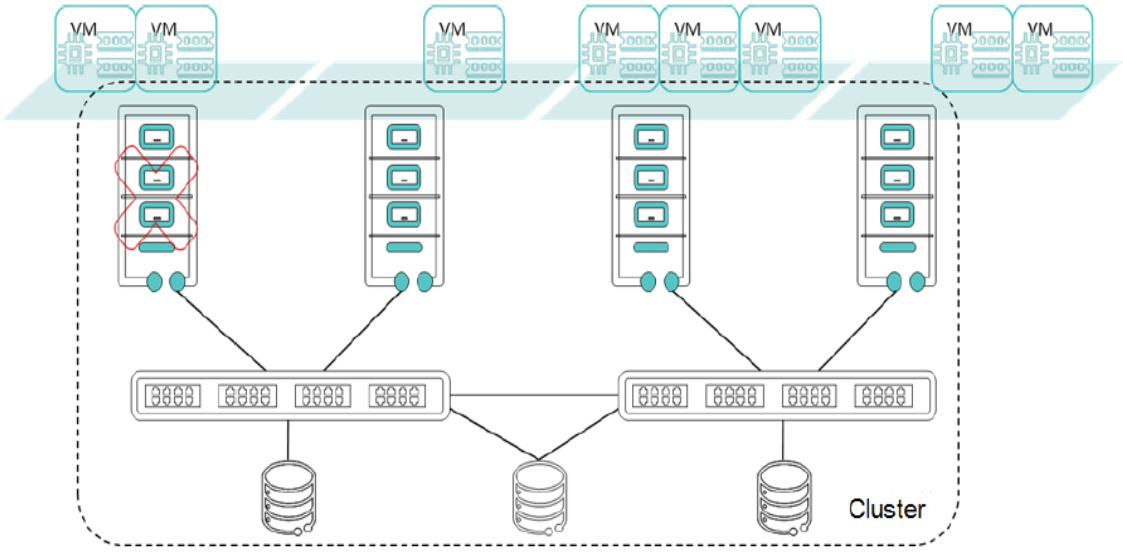
- **Detecting host faults**

  To check whether a compute node is faulty, the administrator needs to periodically establish communication with all nodes in the cluster. If the communication with a node fails, the node may be faulty. Use Huawei FusionCompute as an example. The heartbeat mechanism between the CNA host and VRM enables VRM to check whether CNA is faulty.

- **Handling VM startup failures**

  When a VM is started on another host, services on the VM may fail to automatically start, and even the OS may fail to start. Service recovery on the VM may fail or take a long time.

  In this case, HA on the service plane is needed. If the active VM is faulty or cannot be restored, services will be recovered on the standby VM using HA technologies such as **the floating IP address** and **Keepalived**.



HA Diagram

2. Load Balancing

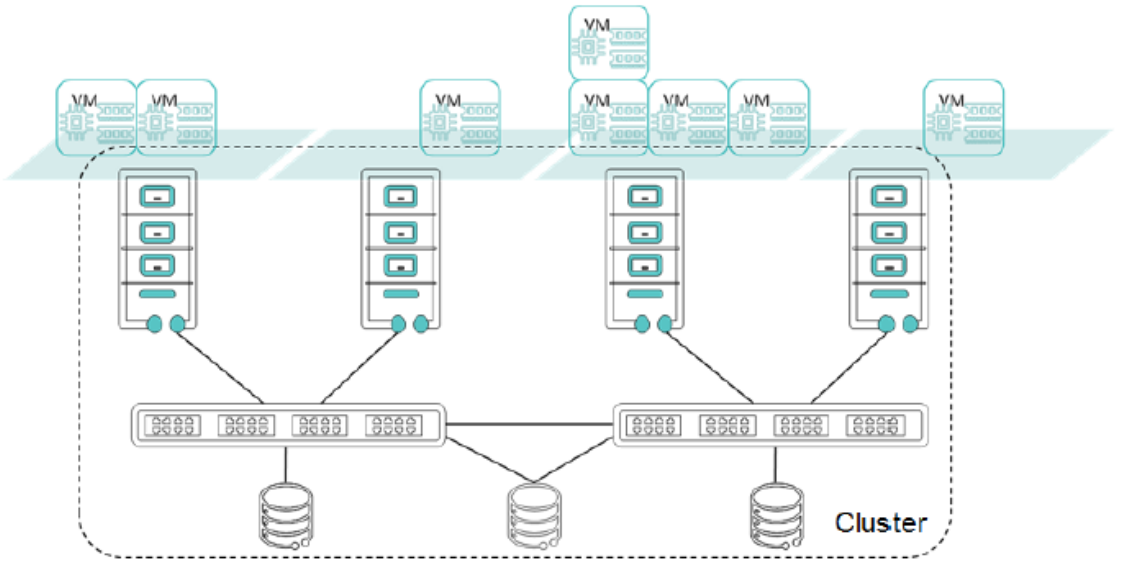
Load balancing is a cluster technology that distributes loads (such as network services and traffic) to multiple network devices (such as servers and firewalls) or links. This improves service processing capabilities and ensures high service reliability.

How Load Balancing is accomplished?

In a virtualization environment, load balancing is generally implemented on compute nodes based on the node CPU and memory usages. During VM creation and running, the management system detects the usage of all physical resources in the cluster, uses the intelligent scheduling algorithm to determine the optimal host for running VMs, and migrates VMs to the optimal host by means of live migration, thereby improving global service experience.

The system administrator can specify the load threshold. In Huawei FusionCompute, if the CPU and memory usage of a host exceeds **60%, VRM** migrates VMs on the host to other hosts.

Before the migration, the administrator can set an **automatic** migration task. Alternatively, the administrator can **manually** perform the migration after receiving a notification.



Load balancing diagram

Load Balancing Advantages

- **High performance:** Load balancing evenly distributes loads to multiple devices, improving the performance of the entire system.
- **Scalability:** Load balancing easily increases the number of devices or links in a cluster to meet the service growth requirements without degrading the service quality.
- **High reliability**: If one or more devices or links are faulty, services are not interrupted, improving system reliability.
- **Easy management.**